

Total No. of Questions : 8]

SEAT No. :

PE-2731

[Total No. of Pages : 2

[6583]-311

T.E. (Mechanical/Electrical)

(302033MJ) HONORS IN E-VEHICLE SYSTEM DESIGN

(2019 Pattern) (Semester - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Four questions from the following.*
- 2) *Draw neat labeled diagrams wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Use of non-programmable electronic calculator is permitted.*
- 5) *Assume Suitable / Standard data if necessary.*

Q1) a) Explain traction control system in EV. **[8]**

b) Classify braking system used in automotive. **[9]**

OR

Q2) a) Explain elements of vehicle dynamics system. **[8]**

b) Explain with neat diagram regenerative braking system (ABS). **[9]**

Q3) a) Explain Powertrain in e-vehicle. **[8]**

b) Explain Two wheel drive layout design. **[9]**

OR

Q4) a) Explain Transmission system in HEV. **[8]**

b) What is Torsen and active differential system. **[9]**

Q5) a) Explain battery compartment and its need in E-vehicle. **[9]**

b) Write short note on: **[9]**

i) Battery life analysis

ii) Pack cooling system

OR

P.T.O.

- Q6)** a) Explain battery performance degradation modelling. [9]
b) Explain with neat diagram Air and Liquid cooling used for battery compartment. [9]

- Q7)** a) Explain crash analysis of EV [9]
b) Explain the importance and process involved in Crash/Impact analysis. [9]

OR

- Q8)** a) What do you mean by Vehicle dynamics? Explain components of vehicle dynamics with neat sketches. [9]
b) Explain Vehicle dynamics and its components. [9]

